



Department of Computer Science

National University of Computer and Emerging Sciences
Lahore Campus

Data Mining

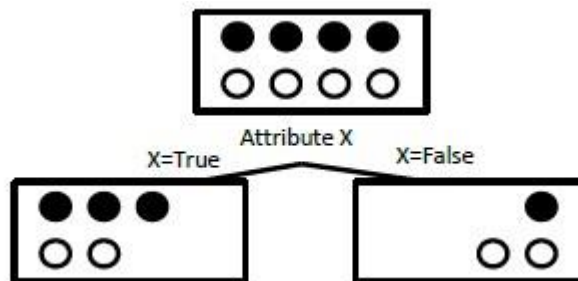
Section A

Quiz 1

Solution

Question 1: 3 marks

Consider the figure below. There are 4 black and 4 white balls, initially evenly mixed. At a branch node, a Boolean attribute X splits the balls into two subgroups with uneven distributions, as shown in the figure.



Compute the information gain for splitting on attribute X above.

$$H(Y) = H(4/8) = 1$$

$$P(X=T) = 5/8 \quad P(Y=\text{black}|X=T) = 3/5$$

$$H(Y|X=T) = -3/5 * \log(3/5) - 2/5 * \log(2/5) = 0.97$$

$$P(X=F) = 3/8 \quad P(Y=\text{black}|X=F) = 1/3$$

$$H(Y|X=F) = -1/3 * \log(1/3) - 2/3 * \log(2/3) = 0.92$$

$$\begin{aligned} IG &= H(Y) - H(Y|X) \\ &= H(Y) - [P(X=T) H(Y|X=T) + P(X=F) H(Y|X=F)] \\ &= 1 - (5/8 * 0.97 + 3/8 * 0.92) = 0.04875 \end{aligned}$$

Question 2: 5 marks

A	B	C	P(A,B,C)
F	F	F	0.2
F	F	T	0.1
F	T	F	0.04
T	F	F	0.06
T	T	F	0.3
F	T	T	0.01
T	F	T	0.09
T	T	T	0.2

- 1) What is $P(A=T|C=T)$
- 2) What is $P(A=T|B=T)$

1. What is $P(A=T|C=T) = .725$
2. What is $P(A=T|B=T) = .905$

Q 3: 2 marks

True/False

- Any probability query (marginal, posterior condition probability) can be answered from the full joint probability distribution using product, sum, and Bayes' rules. **True**
- Algorithms C4.5, which uses information gain ratio as a splitting criterion has selection bias towards features with a greater number of distinct values. **False**